जम्मू केंद्रीय विश्वविद्यालय

Central University of Jammu

रह्या-सुचानी (बगला), जिला सांबा -181143, (जम्मू और कश्मीर)

Rahya-Suchani (Bagla), District: Samba – 181143, Jammu (J&K)

16th July 2025

Roll No. [redacted] B E C S E [redacted]

SEMESTER END-TERM EXAMINATIONS

Programme	:	**B. Tech. CSE & CSE-Cyber**	**Semester**	:	**VI**
Course Name	:	**Deep Learning**	**Course Code**	:	**BECSE3C008, BECCS3C008**
Duration	:	**3 Hours**	**Credits**	:	**03**
Maximum Marks	:	**75**	**Minimum Marks**	:	

After pursuing this course, the student will be able to:	
CO 01	To understand the fundamental concepts of deep learning and its architecture.
CO 02	Design, train and optimize neural networks using advanced techniques.
CO 03	Use regularization and optimization techniques to improve model performance.
CO 04	Implement various deep neural networks such as Convolutional and Recurrent Neural Networks, Transformers etc.
CO 05	Apply deep learning to solve real-world problems and evaluate model results effectively.

Part A **Very Short Answer Type Logical Questions of 03 marks related to different COs**		**Total Marks 3x5=15**	
S. No.	**Statement**	**Levels**	**CO**
A1.	Compute the Cross Entropy Loss between the actual values (y) and predicted values ($\hat{y}$) for the ReLU neuron, given the input vector X= [1, 3, 2, 1], output vector O = [0.7] and weight vector [0.3, 0.6, 0.5, 0.1].	L1	CO-01 CO-02
A2	Explain how LSTMs solve the Vanishing gradient problem?	L2	CO-04
A3.	What is the role of the learning rate in the Gradient Descent algorithm?	L1	CO-02 CO-03
A4.	What is overfitting and underfitting. and how it can be detected?	L2	CO-03
A5.	Solve the XOR problem using a Multi-Layer Perceptron (MLP) and the brute-force optimization technique.	L4	CO-01 CO-03 C0-04

PART B **Short Answer Type Analytical Questions of 6 marks related to different COs**		**Total Marks 6x5=30**	
B1.	Explain the significance of convolution and pooling operations in a CNN with a suitable example?	L4	CO-04
B2.	a) What are the drawbacks of the Sigmoid activation function in deep neural networks? b) Derive how gradient can be computed as a function of the predicted value, actual value and corresponding input in a single neuron.	L3	CO-03 CO-04

B3.	What is regularization? What are the different regularization techniques used for optimizing deep neural networks?	L5	CO-02
B4.	Design a deep convolution neural network for image classification using TensorFlow and Keras library. The architecture comprises of the following: Convolution layer > max-pool layer > drop-out layer, and a fully connected network.	L2	CO-04 CO-05
B5.	Discuss the mathematical equations and intuition behind following optimization algorithms? a) RMS Prop b) AdaGrad c) Adam Compare their strengths and weaknesses, and explain their impact on deep learning model performance.	L3	CO-01 CO-02
	Part C **Long Answer Type Descriptive Questions of 15 marks related to different COs** **(Only Two questions to be attempted)**	**Total Marks 15x2=30**	
C1.	Compare Vanilla RNN, LSTM, and GRU in terms of architecture, advantages, and disadvantages.	L4	CO-04 CO-05
C2.	a) Compare and contrast AlexNet and VGGNet in terms of architecture and performance. b) Consider a CNN with the following architrecture: • Convolution layer:16 filter of 3 x 3 size, stride 1, no padding • Max-pooling layer: 1 filter of 2 x 2 size, stride 2 • Convolution layer: 32 filter of 3 x 3 size, stride 1, padding 0 • Max-pooling layer: 1 filter of 2 x 2 size, stride 2 • Fully Connected Layer: 128 neurons • Output layer: 10 neurons Calculate the total number of trainable parameters in the network?	L3	CO-03 CO-04
C3.	a) Explain the architecture of the Transformer model in detail, focusing on multi-headed, self-attention, and positional encoding? b) Consider a simple neural network with one hidden layer with 3 neurons where you can use Sigmoid function as the activation function and Mean Square Error (MSE) as the loss function and input layers comprises of the two nodes, X1 and X2 where X1 = 1.8 and X2 = 2.1 and target output, y = 0.5. Initialize weights and bias between -1 and 1. Perform two steps of back-propagation and update the weights and bias values accordingly?	L4	CO-01 CO-02